Exam 2 6

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Question 1:

Proof. By definition of even/odd, for some integers a, b, we have $x = 2a$ and $y = 2b + 1$. Then,

$$\begin{aligned} z &= (2a)^2 + (2b+1)^2 \\ &= 4a^2 + 4b^2 + 4b + 1 \\ &= 4k + 1, \text{ where } k = a^2 + b^2 + b. \end{aligned}$$

So k is an integer. Then $z \in R$, and so $L \subseteq R$.

Consider $k = -1$. Then, $4k + 1 = -3 \in R$. But, L is the sum of squares, so all elements in L are positive. Therefore, $R \not\subseteq L$. □

Question 2:

(a)

Proof. Let $S = (0, 1)$. For sake of contradiction, assume that S has a greatest element, say x. Since $x \in S$, $0 < x < 1$.

Consider $y = \frac{x+1}{2} < \frac{1+1}{2} < 1$. Then, $y \in S$. But, $y > \frac{x+x}{2} = x$. This is a contradiction since $y > x$. Thus, S does not have a largest element. □

(b)

Proof. Assume that A, B, C, D are sets satisfying $A \subseteq B$ and $C \subseteq D$. Suppose $(x, y) \in A \times C$. Then, $x \in A$ and $y \in C$. □

Question 3:

Proof. Let $S = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29\}$. Then, we partition it into $P = \{\{3, 9, 27\}, \{5, 15, 25\}, \{7, 21\}, \{1\}, \{11\}, \{13\}, \{17\}, \{19\}, \{23\}, \{29\}\}$. Let these subsets of S be the pigeonholes, and the 11 selected numbers be the pigeons. Then, by the pigeonhole principle, there 2 of the selected numbers must be in the same partition of S. Since all elements of the partitions of S with more than 1 element are divisible by each other, we have found our pair. □

Question 4: Question 6: